

LECTURE ACTIVITY 5: CAN DOGS UNDERSTAND HUMAN CUES? – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Implement the six steps of a statistical investigation given a particular statistical study
- 2) Explain and apply the 3S Strategy (Statistic, Simulate, and Strength of evidence) when evaluating claims about a single population proportion using the count or sample proportion as the statistic
- 3) Specify an appropriate chance model for a given “set scenario” or for a given “set of assumptions”
- 4) Describe and conduct a simulation with many repetitions under the chance model by hand and using the One Proportion applet
- 5) Interpret the results from a simulation
 - Identify what a dot on the plot is in the context of the study.
 - Identify whether the observed statistic is unlikely under the assumption that the chance model is true.
 - Identify the strength of the evidence against the chance model
 - Identify whether the results of the study are unlikely to occur just by random chance.
- 6) Use the results of the simulation to answer the research question in context.

Key Terms:

- Six Step Method
 - 3S Strategy
 - Chance Model
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Introduction

Dogs have been domesticated for about 14,000 years. In that time, have they been able to develop an understanding of human gestures such as pointing or glancing? How about similar nonhuman cues? Researchers Udell, Giglio, and Wynne tested a small number of dogs in order to answer these questions. In this exploration, we will first see **whether or not dogs can understand human gestures**.

To test this, the researchers positioned the dogs about 2.5 meters from the experimenter. On each side of the experimenter were two cups. The experimenter would perform some sort of gesture (pointing, bowing, looking) toward one of the cups. The researchers would then see whether the dog would go to the cup that was indicated. There were six dogs tested, but we will look at one of the dogs (a four-year-old mixed breed named Harley). Each trial involved the experimenter bowing toward one of the cups to see whether Harley would go to that cup.

Our goal in this Lecture Activity will be to go through the **six steps of a statistical investigation**, as applied to this study. (I hope you remember these from today’s reading!)

Step 1: Ask a Research Question

The research question we will investigate in this activity is: **can Harley understand human cues?**

Step 2: Design a Study and Collect Data

Harley performed 10 trials, and for 9 of those he chose the cup that the experimenter bowed toward.



Question 1

What is the sample size in this study? In other words, how many total trials did Harley perform?

Solution: Harley performed 10 trials, meaning the sample size is **10**.



Question 2

Identify one observational unit in this study. **Hint:** use your answer to the previous question to help you. Remember to respond in singular form! **Another hint:** is our data a sample from a population, or observations of a random process?

Solution: An observational unit is **one trial by Harley**.



Question 3

Identify the variable in the study, and explain why it is categorical. What are the possible levels of this variable, and which level are you considering a “success?” **Hint:** recall that a variable is something we can measure from each observational unit.

Solution: The variable is **whether or not Harley chose the cup the experimenter bowed toward**, which is categorical since its values are either **correct cup** or **incorrect cup**, and not numbers. A “success” is a value of **yes** (or **correct cup**) since we’re trying to track the sample proportion of times Harley *correctly* chooses the cup the experimenter bowed toward.

Step 3: Explore the Data

With categorical data, we typically report the sample number of “successes” (count) or the sample *proportion* of successes as the statistic.



Question 4

Report the observed statistic as a count and as a proportion of successes for Harley’s trials.

Solution: The count is **9**, and the sample proportion is **$9/10 = 0.9 = 90\%$** .

Question 5

Below is a bar graph showing the results of the study (sample distribution). To this plot, add:

- A descriptive title, which should also include some mention of what type of plot this is (remember those from Lecture 2?)
- A descriptive x-axis title
- Descriptive x-axis labels

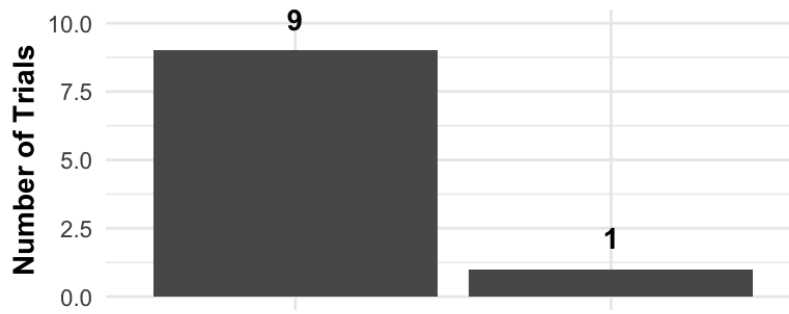


Figure 1: Unlabeled Sample Distribution

Solution:

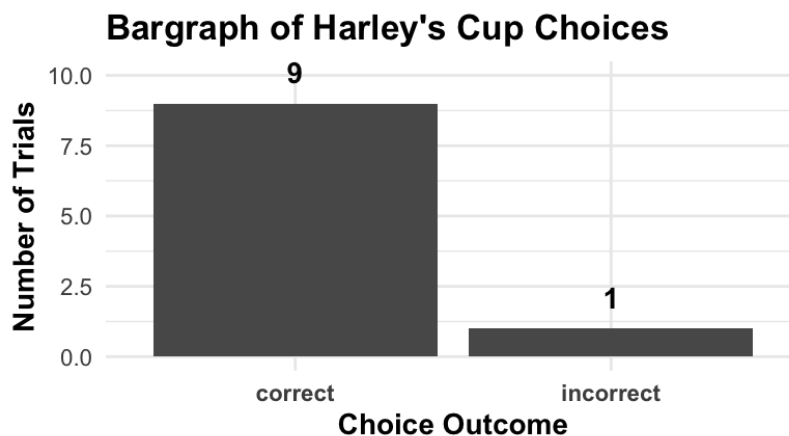


Figure 2: The plot with the desired modifications.

Question 6

Could Harley have gotten 9 out of 10 correct even if he didn't really understand the human gesture and was just randomly guessing between the two cups?

Solution: Yes, it is technically possible to get 9 out of 10 correct when purely guessing.

 Question 7

Do you think it is *likely* that Harley would have gotten 9 out of 10 correct if he was just guessing randomly each time?

Solution: No, it isn't very likely.

Step 4: Draw Inferences Beyond the Data

To reiterate, there are two possibilities for why Harley chose the correct cup 9 out of 10 times:

- **Possibility 1:** He is doing something other than merely guessing and perhaps understands what the experimenters mean when they bow towards the cup.
- **Possibility 2:** He is merely picking a cup at random and in these 10 trials happened to guess correctly in 9 of them. That is, he got more than half correct just by random chance alone.

 Caution

As a (perhaps disappointing spoiler), we will never be 100% certain which of the two possibilities is correct. However, with the statistical tools we'll develop over the next few weeks, we'll be able to say which we think is more *plausible*, given the data we observed.

Notice that the distinction between possibilities 1 and 2 can be succinctly summarized in terms of the following parameter: the long-run frequency at which Harley would choose the correct cup if the scenario were repeated a large number of times. Recall that, by definition, this is just the **probability** that Harley will choose the correct cup. Indeed, because our data is observations from a random process, this is precisely the **parameter** of interest (as we learned in Lecture Activity 4)!

 Question 8

- (a) Under possibility 2 (Harley is just guessing at random), what would be the value of the parameter? Give a specific value.

Solution: If Harley were guessing purely at random, we would expect his long-run proportion of correct guesses to be roughly equal to his long-run proportion of incorrect guesses - that is, we would expect the value of the parameter to be **0.5**.

- (b) Under possibility 1 (Harley is doing something beyond just guessing), what would be the possible range of values for the parameter of interest?

✓ **More than 0.5**

☐ Equal to 0.5

☐ Less than 0.5

Solution: If Harley actually understood human queues, we'd expect his long-run proportion of correct guesses to exceed that which we would expect if he were purely guessing - that is, we'd expect the value of the parameter to be **greater than 0.5**.

Speaking of Lecture Activity 4, recall that our “archetypal” example of a random process is tossing a coin. Indeed, we can try and draw some analogies between tossing a coin and Harley guessing a cup at random.

 Question 9

Explain how we could use a coin toss to represent Harley's choices if he is guessing between the two cups each time. Make sure you state (1) how many times you have to flip the coin to represent one set of Harley's attempts (**hint:** remember the data we have) and (2) what the coin landing "heads" represents in the context of Harley's guesses.

Solution: Flip a coin 10 times to represent Harley's 10 trials. Let heads = correct cup and tails = incorrect cup.

Equipped with this analogy, let's actually carry out this set of coin tosses!

 Question 10

Come up to the front of the room and pick up a coin from me (Dr. Ethan). Then, toss this coin 10 times (why 10?) and count the number of heads you observe. Write down your number of heads below, and add a dot to the dotplot on the whiteboard at the front of the classroom.

I observed _____ heads among my 10 tosses.

Here is a dotplot summarizing the results from across my three lecture sections (60 responses in all). Kindly note that 33 observations were collected from physical coin tosses; the remainder were results from the One Proportion Applet (this disparity was due to the fact that later sections were moved to Zoom, due to heat advisories.)

Dotplot of Number of Heads

Out of 10 Tosses

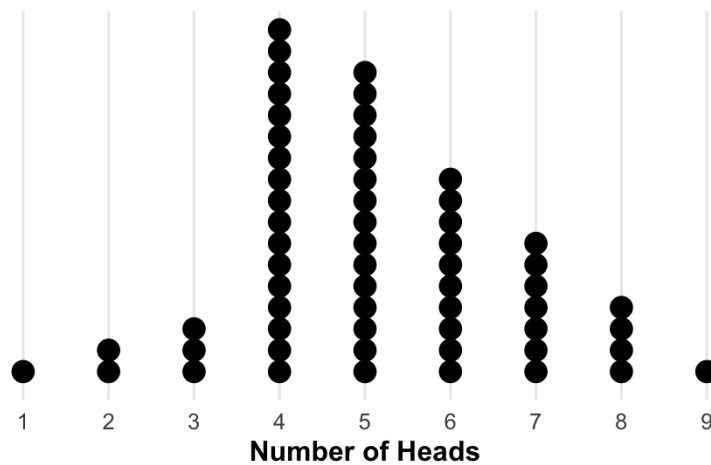


Figure 3: Collective Dotplot from across Dr. Ethan's three STAT 1110 Sections.



Question 11

- (a) Using the context of the problem, explain what a dot on our collective dotplot represents.

Solution: One dot represents one simulated set of 10 trials, showing how many times Harley chose correctly if he was guessing.

- (b) Where does 9 heads fall in the distribution? Would you consider it an unusual outcome, or a fairly typical outcome for the number of heads in 10 flips?

Solution: 9 heads is far in the right tail. It is unusual to see 9 heads out of 10 if we assume a 50% chance of heads.

- (c) Based on your answer to the previous part, do you think it is plausible that Harley was just guessing which cup to choose? Explain briefly.

Solution: No, I do not think it is plausible that Harley was just guessing. The result of 9 out of 10 was an unusual result in the distribution which assumes Harley was just guessing between the cups, so it would be unlikely to see 9 correct trials out of 10 just by chance alone. There seems to be something else going on.



Pro Tip

Congratulations - we just carried out a simulation by hand (without computers)!

Now that we have a general idea of which possibility we think is more likely, let's see how a statistician would think about the problem.

The Chance Model

As we have alluded to in the questions above, the data in this study comes to us in the form of observations from a random process. Hopefully you're starting to see how such data - specifically, statistics resulting from such data - can be used to determine whether the observed statistic is consistent with the values of the statistic simulated by the **chance model** (i.e. a representation of the underlying scenario as a random process).

In an ideal world, we'd gather many more samples. This amounts to replicating the study many times; i.e. giving Harley another set of 10 cues and checking the proportion of times he chooses the correct cup, and then doing it again for another set of 10 cues, and so on and so forth until we end up with a bunch of different sample statistics.

Why would we do this? Well, to be able to better assess whether Harley's results are consistent with the results from the chance model. If we determine that Harley's results are *not* consistent with the results from the random chance model (i.e. that Harley's results are unlikely to have occurred by chance alone), we'll consider this to be evidence *against* the chance model and in favor of our research question (that he understands the bowing gesture).

Of course, we can't perform the actual study more times in order to assess whether Harley is doing something more than just guessing, but we can use *simulations*, just like we did with the coin tosses above!

Applet: Simulate Flipping a Coin Many Times

We each performed one additional simulated sample from the study above, when we each tossed a coin 10 times and recorded the sample proportion of heads observed. However, if we want to repeat this many more times, we'd probably like to rely on technology a bit more.

Thankfully, we have yet another applet to help us! In particular, navigate to the [One Proportion Applet](https://www.rossmanchance.com/applets/2021/oneprop/OneProp.htm) (for those of you with a printed copy of this handout, here is the url: <https://www.rossmanchance.com/applets/2021/oneprop/OneProp.htm>).

Question 12

- (a) In the applet, set the “Number of Tosses” to be 10, and leave the other options as-is. Click on the “Draw Samples” button, and make note of your resulting number of heads below.

- I observed 8 heads among my 10 tosses.

- (b) Why did we leave the value of “probability of heads” as 0.5?

Solution: Setting the probability of heads to be 0.5 corresponds to Harley guessing between the two cups.

- (c) Uncheck the “Animate” button, and press the “Draw Samples” button 9 more times. This will demonstrate how the number of heads varies randomly across each set of 10 tosses; nine more dots should appear on your dotplot as well. Is a pattern starting to emerge? If so, what is that pattern?

Solution: Your dotplot will likely look slightly different from mine, but here's what mine looked like:

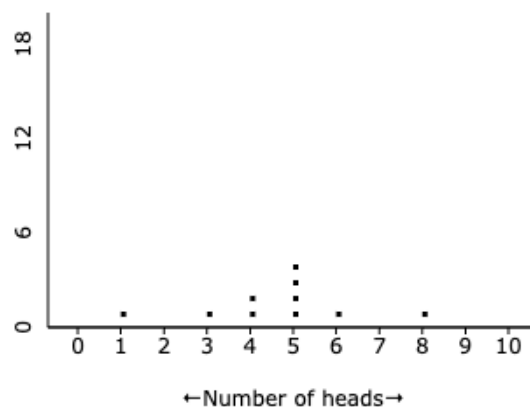


Figure 4: Dotplot from the applet

In particular, note how there are many more points near 5, and fewer points in the tails.

- (d) Now change the “Number of Repetitions” from 1 to 9,990, and press “Draw Samples” again. The applet will now show the result for the number of heads in 10,000 different sets of 10-coin tosses. So, each dot represents the number of times Harley chooses the correct cup out of 10 attempts assuming he is just guessing. Insert a screenshot of your dotplot (or a sketch of your

dotplot) below.

Solution:

Again, your specific plot will likely look a little different from mine (because of sampling variability), but it should look something like the following:

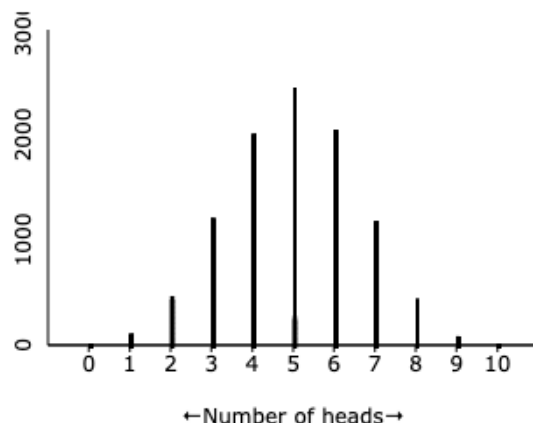


Figure 5: Dotplot from the applet

Indeed, our pattern has become even more apparent- values near 5 are the most common, whereas values in the tails are least common.

Let's take a moment to remember *why* we conducted this simulation: to assess whether Harley's results (9 correct in 10 attempts) would be unlikely to occur by chance alone if he were just guessing between the pair of cups for each attempt.



Question 13

- (a) In your dotplot above, locate the result of getting 9 heads. Would you consider this an unlikely result in the tail of the distribution of the number of heads, or a likely one near the middle?

Solution: Seems like a pretty unlikely result! In particular, a value of 9 is all the way in the right tail of the dotplot, where points are less common.

- (b) Based on the results of the 10,000 simulated sets of 10-coin flips each, would you conclude that Harley would be very unlikely to have picked the correct cup 9 times in 10 attempts if he was randomly guessing between the two cups each time? Explain how your answer relates to the plot from Question (12) above.

Solution: Yes. Based on the simulated data in the dotplot, it was very unlikely to observe 9 out of 10 under the assumption that Harley was just randomly guessing. We can see that these values rarely appear on the plot, therefore, we can conclude that Harley would be unlikely to get 9/10 correct if he was just guessing.

- (c) Do the results of this study appear to be unlikely to have occurred by chance alone? Explain.

Solution: Yes. The result appears unlikely to have occurred by chance alone since it was very unlikely in the distribution.

- (d) Is it plausible that Harley was just guessing?

Solution: No. It is not plausible that Harley was just guessing; there seems to be something else going on.

Summarizing Your Understanding

To make sure you understand the coin-flipping chance model, examine the information below which indicates what parts of the real study correspond to the physical (coin-flipping) simulation.

One Coin Flip	= One of Harley's Attempts
Coin Landing Heads	= Harley picking the correct cup
Tails	= Harley picking the incorrect cup
Chance of Heads	= 0.5 (long-run proportion of Harley picking the correct cup if he is just guessing)
One Repetition	= One set of 10 simulated attempts by Harley

The 3S Strategy

We will call the process of simulated could-have-been statistics under a specific chance model the **3S Strategy**. After forming our research conjecture and collecting the sample data, we will use the 3S strategy to weight the evidence against the chance model. This 3S strategy will serve as the foundation for addressing the question of Statistical Significance in Step 4 of the statistical investigation method.

- **Statistic:** Compute the value of the statistic from the observed sample data
 - In this study, the value of the statistic is $9/10 = 0.9$
- **Simulate:** Identify a “by-chance-alone” explanation for the data. Repeatedly simulate values of the statistic that could have happened when the chance model is true.
 - In this study, we flipped a coin 10 times and kept track of how many times it came up heads. We then repeated this process 9999 more times, each time keeping track of how many heads were obtained in each of the 10 flips.
- **Strength of Evidence:** Consider whether the value of the observed statistic from the research study is unlikely to occur if the chance model is true. If we decide the observed statistic is unlikely to occur by chance alone, then we can conclude that the observed data provide strong evidence against the plausibility of the chance model. If not, then we consider the chance model to be a plausible (believable) explanation for the observed data; in other words, what we observed could plausibly have happened just by random chance.
 - In this study, because we rarely obtained a value of 9 heads when flipping the coin 10 times, this means that it is unlikely that Harley is just guessing, because if Harley was just guessing he rarely would get a value like 9 correct out of 10 attempts.

Step 5: Formulate Conclusions

Question 14

- (a) Based on your previous answers, do you have evidence to say that the long-run proportion that Harley will choose the correct cup is more than 0.50? In other words, do you have evidence to say that Harley will choose the correct cup more than half the time in the long run?

Solution: Yes. Since 9 out of 10 is unusually high under random guessing, there is evidence that Harley's long-run proportion of choosing correctly is more than 0.50.

- (b) Based on this analysis, are you convinced that Harley can understand human cues? Why or why not?

Solution: This study gives evidence that it was unlikely to see Harley's results of 9 out of 10 correct trials if we assume that Harley was just guessing. So, I do not think that Harley was just guessing. It might be that Harley can understand human cues.

Step 6: Look Back and Ahead

Question 15

A single study will not provide all of the information needed to fully understand a broad, complex research question. Thinking back to the original research question, what additional studies would you suggest conducting next?

Solution: Some suggestions include testing more dogs, using more trials, testing different human cues, and comparing human cues with nonhuman cues. One student also recommended testing to see if there were differences between species of dogs.

Take It Further - Optional

One important step in a statistical investigation is to consider other models and whether the results can be confirmed in other settings.

In a different study, the researchers used a mechanical arm (roughly the size of a human arm) to point at one of the two cups. The researchers tested this to see whether dogs understood nonhuman gestures. In 10 trials, Harley chose the correct cup 6 times.

- 1) Using the dotplot you obtained when you simulated 10,000 sets of 10 coin flips assuming Harley was just guessing, locate the result of getting 6 heads. Would you consider this an unlikely result in the tail of the distribution?

Getting 6 out of 10 is not unusual; it is close to the center of the distribution.

- 2) Based on the results of 10,000 simulated sets of 10 coin flips each, would you conclude that Harley would be very unlikely to have picked the correct cup 6 times in 10 attempts if he was randomly guessing between the two cups each time? Explain how your answer relates to the applet's dotplot.

No. Six correct choices could easily happen by chance alone. We can see this because the value of 6 is very likely in the distribution.

3) Is this study's result unlikely to have happened by chance?

No.

4) Do the results of this study suggest that Harley just guessing is a plausible explanation for Harley picking the correct cup 6 out of 10 times?

Yes. Guessing is a plausible explanation for 6 out of 10 correct.

5) Does this study prove that Harley cannot understand the mechanical arm?

No. It does not prove Harley cannot understand the mechanical arm.

Compare the analyses between the two studies. How does the unusualness of the observed statistic compare between the two studies? Does this make sense based on the value of the observed statistic in the two studies? Does this make sense based on how the two studies were designed? Explain. Hint: Why might the results differ for human and mechanical arms? Why would this matter?

The 9 out of 10 result is much more unusual than 6 out of 10. This makes sense because 9 is far from the guessing expectation of 5, while 6 is close to 5. The results may differ because dogs may understand human gestures better than mechanical-arm gestures.